Module 7 – Reporting & Power Platform

User Story 8: Operational Reporting

Operational Reporting (Non-SSRS) — Appointment Schedules, No-Show Rates, Visit Volumes

Project: HMS / Folder: `documentaion/ssrs`

Purpose
- Produce normal (non-SSRS) reports and diagrams that satisfy the acceptance criteria in User Story 8.

Acceptance Criteria mapping
- Parameterized reports: describe parameters and example usage (clinic, date range, provider).
- Clear formatting and grouping: show example layouts and grouping rules.
- Accurate mapping to SQL data sources: include SQL queries and table/column mappings.

Report 1 — Appointment Schedule (parameterized)
- Parameters:
  - `@ClinicID` (int, optional) — filter by clinic
  - `@StartDate` (date) — report start date
  - `@EndDate` (date) — report end date
  - `@ProviderID` (int, optional) — filter by provider

```
-- Sample SQL (for relational DB like SQL Server)
SELECT
  a.AppointmentID,
  a.AppointmentDateTime,
  a.Status, -- Scheduled / Completed / NoShow / Cancelled
  p.PatientID,
  p.FullName AS PatientName,
  pr.ProviderID,
  pr.FullName AS ProviderName,
  c.ClinicID,
  c.Name AS ClinicName
FROM Appointments a
JOIN Patients p ON a.PatientID = p.PatientID
JOIN Providers pr ON a.ProviderID = pr.ProviderID
JOIN Clinics c ON a.ClinicID = c.ClinicID
WHERE a.AppointmentDateTime >= @StartDate
  AND a.AppointmentDateTime < DATEADD(day, 1, @EndDate)
  AND (@ClinicID IS NULL OR a.ClinicID = @ClinicID)
  AND (@ProviderID IS NULL OR a.ProviderID = @ProviderID)
```

ORDER BY a.AppointmentDateTime;

Formatting & Grouping
- Group by `ClinicName` then `ProviderName` then date (AppointmentDate).
- Show totals per provider (count of appointments) and clinic.
- Columns: Time, PatientName, Status, AppointmentID.

Report layout example (text table)
Clinic: Westside Clinic
Provider: Dr. A. Kumar
Date: 2026-02-01

Time	Patient Name	Status	Appt ID
09:00 AM	John Smith	Scheduled	12345
09:15 AM	Maria Lopez	NoShow	12346
09:30 AM	–	Cancelled	12347

Provider totals: 3 appts (1 no-show, 1 cancelled)

Report 2 — No-Show Rates (parameterized)
- Parameters: `@ClinicID`, `@StartDate`, `@EndDate`, `@Granularity` ('day'|'week'|'month')

```
-- Sample SQL to compute no-show rate by date
SELECT
  CAST(a.AppointmentDateTime AS date) AS ReportDate,
  COUNT(*) AS TotalAppointments,
  SUM(CASE WHEN a.Status = 'NoShow' THEN 1 ELSE 0 END) AS NoShows,
  1.0 * SUM(CASE WHEN a.Status = 'NoShow' THEN 1 ELSE 0 END) / COUNT(*) AS
NoShowRate
FROM Appointments a
WHERE a.AppointmentDateTime >= @StartDate
  AND a.AppointmentDateTime < DATEADD(day, 1, @EndDate)
  AND (@ClinicID IS NULL OR a.ClinicID = @ClinicID)
GROUP BY CAST(a.AppointmentDateTime AS date)
ORDER BY ReportDate;
```

Formatting & Visuals
- Show a small chart (line or bar) of NoShowRate over time (per day/week/month).

- Include a summary table with totals and percentage.

Report 3 — Visit Volumes (parameterized)
- Parameters: `@ClinicID`, `@StartDate`, `@EndDate`, `@GroupBy` (e.g., Clinic/Provider/Department)

```
-- Sample SQL to get visit volumes by provider
SELECT
  pr.ProviderID,
  pr.FullName AS ProviderName,
  COUNT(*) AS VisitCount
FROM Appointments a
JOIN Providers pr ON a.ProviderID = pr.ProviderID
WHERE a.Status = 'Completed'
  AND a.AppointmentDateTime BETWEEN @StartDate AND @EndDate
  AND (@ClinicID IS NULL OR a.ClinicID = @ClinicID)
GROUP BY pr.ProviderID, pr.FullName
ORDER BY VisitCount DESC;
```

Data Mapping (example schema)
- Appointments (AppointmentID PK, PatientID FK, ProviderID FK, ClinicID FK, AppointmentDateTime, Status)
- Patients (PatientID PK, FullName, DOB, Sex)
- Providers (ProviderID PK, FullName, Specialty)
- Clinics (ClinicID PK, Name, Location)

How to represent parameters in a normal (non-SSRS) report
- Provide a small parameters section at top of the document or a form UI in the application.
- Example parameter block:
  - Clinic: "Westside Clinic" (All)
  - Start date: 2026-02-01
  - End date: 2026-02-28
  - Provider: "All"

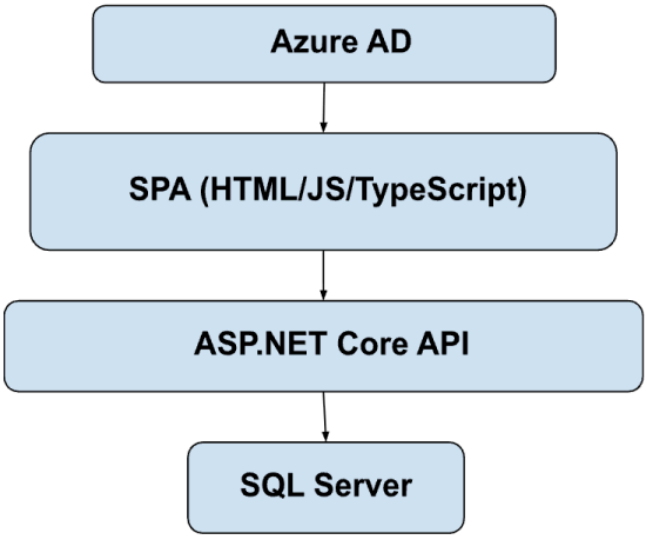
Diagrams

1) Data flow (ASCII)

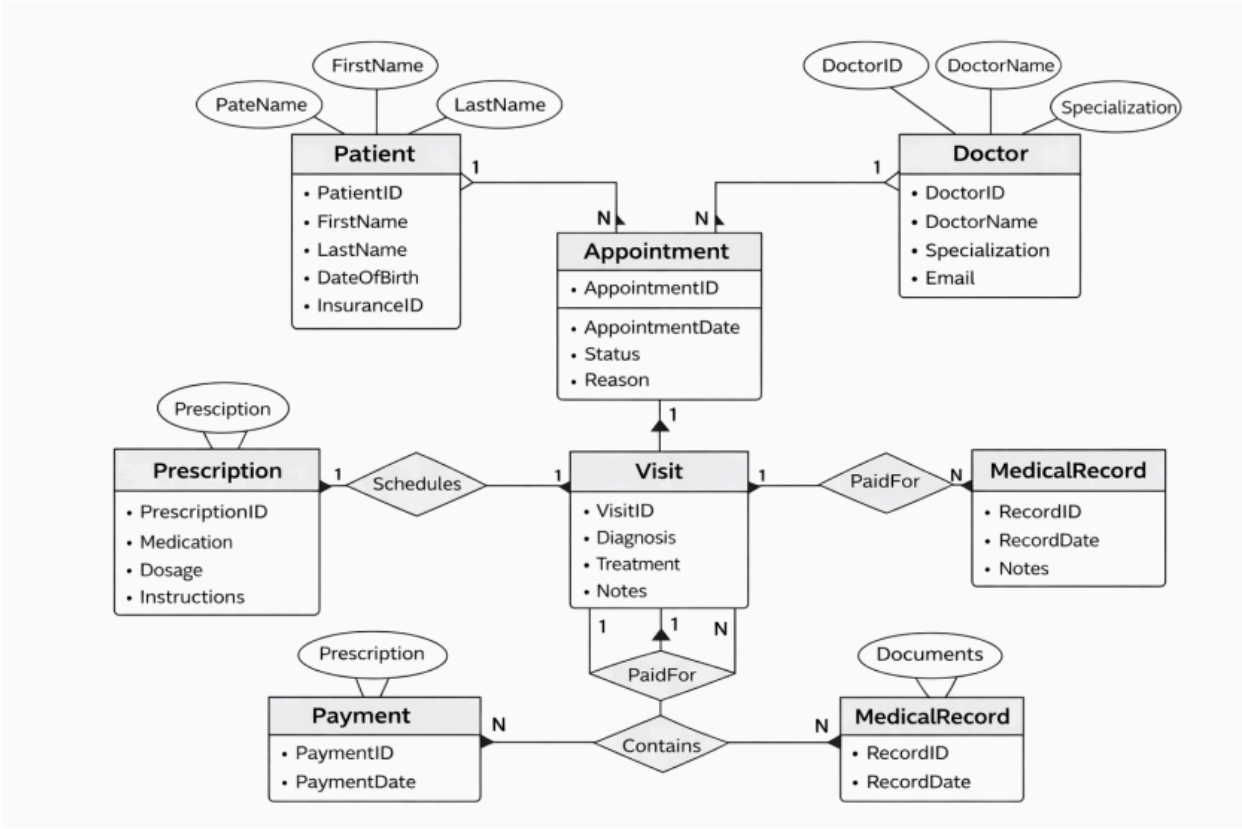
```
  [Operational DB]
        |
        |  (SQL queries shown above)
        v
  [Report generator script / app] ---> [Table / CSV / PDF outputs]
        |
```

v

[Charts: No-show rate, Visit volume]



2) Simplified ER (ASCII)



Mock sample (small table) — No-show summary by clinic

Clinic Name	Total Appts	NoShows	NoShowRate
Westside Clinic	120	12	10.0%
Downtown Clinic	150	30	20.0%

Notes for delivery
- This file is a normal documentation/report file and does not require SSRS.
- To convert to SSRS later: create parameter objects matching `@StartDate`, `@EndDate`, `@ClinicID`, `@ProviderID`; use dataset queries above; define groups and tablix layouts to match the grouping rules.